RYERSON UNIVERSITY

ECN 801 PRINCIPLES OF ENGINEERING ECONOMICS

INSTRUCTOR: TSOGBADRAL GALAABAATAR

TIME: 2 HOURS

DATE: 14 MARCH 2012

Instructions:

1. Choose the option closest to your answer and record the letter you have chosen on the scanner sheet. Use the space after each question to show your work. Answer all 25 questions.

2. No references (formula sheet, notes, books, etc.) are permitted.

3. Calculators that can store and display files are not allowed. Other calculators are allowed.

4. Useful formulas:

Equal Payment Series Present Worth: $P = A\left[\frac{(1+i)^N - 1}{i(1+i)^N}\right]$

Linear Gradient Series Present Worth: $P = G\left[\frac{(1+i)^N - iN - 1}{i^2(1+i)^N}\right]$

Geometric Gradient Series Present Worth: $P = A_1\left[\frac{1-(1+g)^N(1+i)^{-N}}{i-g}\right]$ if $g \neq i$

STUDENT'S NAME:_____

(Please Print)

ID# _____

Good Luck!

1. Suppose 3 years ago, you paid $950 for a bond that has a par value of $1,000. The bond interest is paid annually and annual coupon rate is 4.5%. If you want to have 5% yield on your investment, what should be your selling price today?

a	854.34	**b**	878.31	**c**	1,135.00	**d**	921.45	**e**	957.93

The annual interest payment is $1000×0.045=$45.00.
Solving equation 950=45(P/A,5%,3)+X(P/F,5%,3) for X, we get X=$957.93.

2. Suppose 5 years ago, you purchased a corporate bond with 10-year maturity and a $1,000 par value. The coupon rate is 8% (compounded semi-annually) and bond interest is paid semi-annually. (You received 10 interest payments till now). Suppose the current market interest rate is 10% (compounded semi-annually). What is the current market value of the bond?

a	1,400.00	**b**	1,208.19	**c**	922.77	**d**	989.14	**e**	1,059.67

Semi-annual effective interest rate is 4%. Semi-annual interest payment is $1000×0.04=$40.00. Current market semi-annual (effective) interest rate is 5%. Also, there are 5 (=10-5) years left till bond matures, which means you'll receive 10 interest payments. The current market value is 40(P/A,5%,10)+1000(P/F,5%,10)=922.77.

3. Net cash flows (A_n) and project balance of an investment with a service life of 3 years is given in the following table. Two cash flow amounts are not given.

n	Net Cash Flow (A_n)	Project Balance
0	-4,000	-4,000
1	( )	-1,500
2	2,495	920
3	( )	2,000

Using the given numbers, find the interest rate used in computing the project balance.

a	20%	**b**	25%	**c**	5%	**d**	10%	**e**	15%

Using period 1 and 2, we have the following equation -1,500(1+i)+2,495=920

Solving for I, we have i=0.05.

4. Net cash flows (A_n) and project balance of an investment with a service life of 4 years is given in the following table.

n	Net cash flow (A_n)	Project Balance
0	-20,000	-20,000
1	5,000	( )
2	6,000	-12,700
3	10,000	( )
4	( )	10,000

Find the net cash flow in period 4 (A_4).

a	15,127	**b**	13,280	**c**	12,986	**d**	14,367	**e**	12,700

The project balance at the end of period 1 is -20,000(1+i)+5,000. Using this value, the project balance at the end period 2 can be expressed as (-20,000(1+i)+5,000)×(1+i). We know this should be equal to -12,700. Denote 1+i by x. Then we have the following equation for x
(-20,000x+5,000)x=-12700.
Solving this quadratic equation gives us x_1=1.1 and x_2=-0.85. The corresponding interest rates are i_1=1.1-1=0.1 or i_1=10% and i_2=-0.85-1=-1.85 which can not hold true. Thus, the interest rate used in this calculation is 10%.

We can calculate the project balance at the end of period 3. It is -12,700×1.1+10,000=-3,970. Now we can solve for A_4. -3,970×1.1+ A_4=10,000. Thus, A_4=14,367.

5. Which of the following statement is correct for an investment with one time cash outflow (initial investment $A_0<0$) and cash inflows starting period 1? (Assume interest rate is positive.)
   a) The discounted payback period is equal to the payback period.
   b) The discounted payback period is larger or equal to the payback period.
   c) The discounted payback period is smaller or equal to the payback period.
   d) The discounted payback period is larger than the payback period.
   e) The discounted payback period is smaller than the payback period.\

   The correct answer is b).
   For any $A_n>0$, $A_n/(1+i)^n<A_n$. This means it will take usually more time to recover the initial investment in discounted payback period. But the two periods can be equal. For example, if A_0=-1,000 and A_1=2,000 and i=10%, then both the payback period and discounted payback period is 1.

6. A project's cash flow is given below.

n	Net cash flow (A_n)
0	-5,000
1	1,000
2	3,000
3	3,000
4	1,500
5	1,000

What is the discounted payback period if *i*=25%.

a	3 years	**b**	4 years	**c**	5 years	**d**	No discounted payback period

The present worth of first 4 inflows is, PW_4=4,870.40. The present worth of first 5 inflows is, PW_5=5,198.08>5,000. Thus, the discounted payback period is 5.

7. Cash flows of two independent projects are given below.

n	Project A	Project B
0	-5,000	-6,000
1	1,000	600
2	1,000	900
3	1,000	1,200
4	1,000	1,500
5	1,000	1,800
6	1,000	2,100

Based on Present Worth criteria, what can we say about acceptability of these projects given *i*=5%.

a	Accept both A and B	**b**	Accept A and reject B	**c**	Accept B and reject A	**d**	Reject both A and B

The present worth of the first project is -5,000+1,000(P/A,5%,6)=75.70>0.

The present worth of the second project is -6,000+600(P/A,5%,6)+300(P/G,5%,6)=635.82>0.

We should accept both (independent) projects.

8. Consider the following project at interest rate 5%. Choose the correct statement about the acceptability of the project.

n	Cash flow
0	-1,300
1	700
2	50
3	100
4	200
5	400
6	800

a) The present worth of this project is a positive number; accept the project.
b) The present worth of this project is a positive number; reject the project.
c) The present worth of this project is a negative number; reject the project.
d) The present worth of this project is a negative number; accept the project.

We see a pattern of geometric gradient series starting period 2. The present worth is

$$-1,300+\frac{700}{1.05}+50\times\frac{1-(2/1.05)^5}{0.05-1}\times(P/F,5\%,1)=573.33>0$$

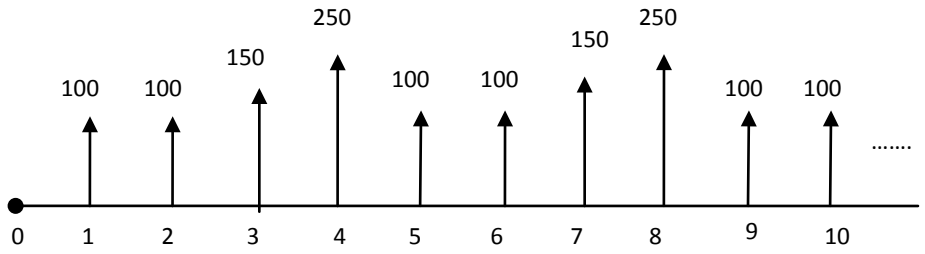
Thus, the correct answer is a)

9. A company opened a bank savings account with *i*=10% for the purpose of donation. Starting next year, the company wants to donate $10,000 to a local church each year. Suppose company never goes bankrupt and they want to continue this donation forever. How much should they deposit today if they want to cover their donation by yearly withdrawal from the account?

a	200,000	**b**	100,000	**c**	150,000	**d**	110,000	**e**	120,000

10,000/0.1=100,000.

10. Consider an investment in which a cash flow pattern that repeats itself every 4 years forever. Find the capitalized equivalent amount at interest rate 5%.



a	2,109.67	**b**	1,897.80	**c**	2,654.78	**d**	3,165.78	**e**	2,939.61

The annual equivalent worth of the first phase (first 4 periods) is

$$100+\left(50(P/F,5\%,3)+150(P/F,5\%,4)\right)\times(A/P,5\%,4)=146.98$$

The capitalized equivalent amount is 146.98/0.05=2,939.61

11. Find the IRR (internal rate of return) of the following investment.

n	Net Cash Flows (A_n)
0	-10,000
1	6,500
2	5,750

a	15%	**b**	25%	**c**	35%	**d**	10%	**e**	20%

$$-10,000+\frac{6,500}{1+i}+\frac{5,750}{(1+i)^2}=0$$

Let x=1/(1+i). Then above equation becomes $5,750x^2+6,500x-10,000$. Solving for x, we get

$X_1=20/23$ and $x_2=-2$. The corresponding interest rates are $i_1=(1-x_1)/x_1=3/20=0.15$ and

$i_2=(1-x_2)/x_2=-1.5$. Since I should be greater than -1, we have interest rate of 15%.

12. For the investment project described in the previous problem (Problem 11), if MARR=20%, what would the company do based on IRR criteria?
    a) The company should accept the project.
    b) The company should reject the project.
    c) The company is indifferent about accepting and rejecting.

MARR=20%>15%=IRR, hence the project should be rejected. The correct answer is B.

13. Find the IRR of the following project.

n	Net Cash Flows (A_n)
0	-10,000
1	0
2	0
3	0
4	15,000

a	12%	**b**	10.67%	**c**	11.54%	**d**	9.73%	**e**	12.83%

PW(i)=-10,000+15,000/$(1+i)^4$=0. Thus $(1+i)^4$=1.5. Hence, i=0.1067.

14. Consider the following project. Evaluate the present worth of the project at interest rates 10% and 20%. Then using these evaluations, find the approximate IRR using linear interpolation method.

n	Net Cash Flow (A_n)
0	-5,500
1	2,000
2	1,000
3	3,000
4	2,000

a	18.24%	**b**	17.95%	**c**	16.36%	**d**	13.27%	**e**	14.52%

PW(10%)=764.60>0 and PW(20%)=-438.27<0. By linear interpolation method,

$$\frac{x-10}{20-x}=\frac{764.60}{438.27}=1.7446 \Rightarrow x=16.36\%$$

15. Suppose the purchase price of a machine with service life of 5 years is $5,000. If the capital recovery cost is $1,000, find salvage value given *i*=5%.

a	855.76	**b**	943.71	**c**	1,130.50	**d**	0	**e**	420.40

$$\left(5,000 - S(P/F,5\%,5)\right) \times (A/P,5\%,5) = 1,000$$
$$S = \left(5,000 - 1,000(P/A,5\%,5)\right) \times (F/P,5\%,5) = 855.759$$

16. A company has purchased a machine for $10,000. The machine has a useful life of 3 years and has a salvage value of $2,000. By using this machine, the company will save $4,000 during the first year, $6,000 during the second and $4,000 during the third year. If the machine is expected to be used 3,000 hours during the first year, 2,500 hours during the second and 2,500 hours during the third year. If the interest rate is 5%, find the equivalent net saving per machine hour.

a	$0.52	**b**	$0.61	**c**	$0.73	**d**	$0.97	**e**	$0.23

The present worth of project PW=-10,000+4,000(P/F,5%,1)+ 6,000(P/F,5%,2) +6,000(P/F,5%,3)=4,434.40

Let c is the net saving per machine hour. Then, the project present worth is PW=3,000c(P/F,5%,1)+ 2,500c(P/F,5%,2) +2,500c(P/F,5%,3)=7,284.2c.

Hence, 7,284.2c=4,434.40 and c=0.6087

17. A soft drink company has to decide whether to make bottles in house or to buy. If they choose to buy, the can buy at market price $0.10 per bottle. They need 200,000 bottles each year. If they choose make-in-house option, they need to buy a machine that costs $100,000 with service life of 20 years and a salvage value of $5,000. The maintenance cost of the machine is $4,000 per year. All other additional cost of the make-in-house option is $20,000 per year. What should the company do? Use *i*=5%.
    a) Choose "buy" option.
    b) Choose "make" option.
    c) The company is indifferent between two options.

    The correct answer is A (choose "buy" option). If we calculate only additional cost of the make-in house option per bottle, it equals 20,000/200,000=$0.10 per bottle, which is the market price of bottle. Since there are initial purchase price and maintenance cost for make-in-house option, it is much expensive than "buy" option.

18. Consider mutually exclusive projects. Based on Present Worth criteria, which project should be chosen if i=5%.

Net Cash Flows				
n	Project A	Project B	Project C	Project D
0	-1,000	-1,000	-1,000	-1,000
1	900	-600	1,000	850
2	600	2,200	550	850

a	Project A	**b**	Project B	**c**	Project C	**d**	Project D

PW_A=401.36, PW_B=423.96, PW_C=451.25 and PW_D=580.49. Project D should be selected since it results the highest PW.

19. Based on the Present Worth criteria, a company is indifferent between two mutually exclusive projects. If i=5%, find the value of X.

n	Project A	Project B
0	-2,000	-1,500
1	1200	-1,000
2	1200	X
3	1200	2,000

a	2,078.40	**b**	2,314.15	**c**	2,196.96	**d**	3,067.74	**e**	2,403.12

The present worth of project A, PW_A=-2000+1200(P/A,5%,3)=1267.84. The present worth of project B, PW_B=-724.8+X(P/F,5%,2).

$$PW_A = PW_B \Rightarrow 1,267.84 = -724.8 + X(P/F, 5\%, 2)$$
$$X = 1,992.64(F/P, 5\%, 2) = 2,196.88$$

20. A company is considering two projects with 3 years of service life. The first project needs the initial investment of $150,000 and an annual maintenance cost of $20,000. The second project needs the initial investment of $200,000. If the incremental rate of return (the rate of return of the incremental investment) is 5%, find the annual operating cost for the second project.

a	5,904	**b**	1,639	**c**	12,340	**d**	3,427	**e**	1,277

Suppose the annual operating cost of project II is X. Then the incremental investment II-I is:

n	Project II-I
0	-50,000
1	20,000-X
2	20,000-X
3	20,000-X

Since the incremental rate of return is 5%, PW_{II-I}(5%)=-50,000+(20,000-X)(P/A,5%,3)=0. Solving for X, we get X=1,640.

21. Cash flows of two mutually exclusive projects are given. If the IRR of the incremental investment is 10%, find X.

n	Project A	Project B
0	-3,000	-3,500
1	1,500	1,500
2	1,200	X
3	1,300	1,300

a	2,000	**b**	1,675	**c**	1,920	**d**	1,745	**e**	1,805

The incremental investment, B-A, is:

n	Project B-A
0	-500
1	0
2	X-1,200
3	0

$PW_{B-A}(10\%)=-500+(X-1,200)(P/F,10\%,2)=0$. Thus, $X=500(1.1^2)+1,200=1805$.

22. Let A and B be two mutually exclusive projects. Suppose the present worth of incremental investment B-A is a positive number. Then,
    a) The present worth of A is greater than the present worth of B.
    b) The present worth of B is greater than the present worth of A.
    c) The present worth of A is equal to the present worth of B.

Since $PW_{B-A}(i)=PW_B(i)-PW_A(i)$, $PW_{B-A}(i)>0$ implies $PW_B(i)>PW_A(i)$. The correct answer is B.

23. A company is considering three projects: Project A1, A2 and A3. The internal rate return for these projects are IRR_{A1}=17.62%, IRR_{A2}=16.60% and IRR_{A3}=15.99%. The incremental rate of returns for incremental investments A2-A1 and A3-A2 are 14.16% and 6.66% respectively. If "do nothing" is an available choice and MARR=16.30%, which project would be chosen?
    a) A1
    b) A2
    c) A3
    d) Not enough information is given to answer the question.

    If the company chooses "do nothing", they'll continue earning MARR rate. Thus, since IRR_{A3}<MARR, project A3 should be rejected. Between A1 and A2, since IRR_{A2-A1}=14.16%<16.30%=MARR, A1 should be chosen. The correct answer is A.

24. Consider two mutually exclusive projects A and B with unequal service lives. On the basis of Annual Equivalent Worth method, which project would be selected under infinite (or indefinite) planning horizon and both projects can be repeated in the future. Use *i*=5%. (*Hint*: Use the Lowest Common Multiple of project lives as analysis period.)

n	Project A	Project B
0	-5,000	-3,000
1	3,000	2,000
2	2,000	2,500
3	2,000	

    a) Project A
    b) Project B
    c) Indifferent between A and B.

    The common service period is 6 (the lowest common multiple of 3 and 2). In order words, project A will be repeated once and B twice in 6 year period. But we know, when we use annual equivalent worth method, it is enough to compare annual equivalent worth of first phase of each projects with 6 year life.
    AEW_A=(-5,000+1,000(P/F,5%,1))(A/P,5%,3)+2000=513.72
    AEW_B=(-3,000+500(P/F,5%,2))(A/P,5%,2)+2000=630.49
    Thus, AEW_B>AEW_A. Project B should be selected.

25. Suppose total annual equivalent cost (AEC) is a function of a (engineering) design variable (A) as follows:

$$AEC = 20A + \frac{320}{A}$$

If you want to minimize the AEC, what is the optimal value for A?

a	1	**b**	2	**c**	3	**d**	4	**e**	5

$$AEC = 20A + \frac{320}{A} \geq 2\sqrt{20A}\sqrt{\frac{320}{A}} = 160$$

The equality holds when 20A=320/A or A=4. Thus, AEC is minimized at A=4.

5.0%

	Single Payment		Equal Payment Series				Gradient Series		
N	Compound Amount Factor (F/P,i,N)	Present Worth Factor (P/F,i,N)	Compound Amount Factor (F/A,i,N)	Sinking Fund Factor (A/F,i,N)	Present Worth Factor (P/A,i,N)	Capital Recovery Factor (A/P,i,N)	Gradient Uniform Series (A/G,i,N)	Gradient Present Worth (P/G,i,N)	N
1	1.0500	0.9524	1.0000	1.0000	0.9524	1.0500	0.0000	0.0000	1
2	1.1025	0.9070	2.0500	0.4878	1.8594	0.5378	0.4878	0.9070	2
3	1.1576	0.8638	3.1525	0.3172	2.7232	0.3672	0.9675	2.6347	3
4	1.2155	0.8227	4.3101	0.2320	3.5460	0.2820	1.4391	5.1028	4
5	1.2763	0.7835	5.5256	0.1810	4.3295	0.2310	1.9025	8.2369	5
6	1.3401	0.7462	6.8019	0.1470	5.0757	0.1970	2.3579	11.9680	6
7	1.4071	0.7107	8.1420	0.1228	5.7864	0.1728	2.8052	16.2321	7
8	1.4775	0.6768	9.5491	0.1047	6.4632	0.1547	3.2445	20.9700	8
9	1.5513	0.6446	11.0266	0.0907	7.1078	0.1407	3.6758	26.1268	9
10	1.6289	0.6139	12.5779	0.0795	7.7217	0.1295	4.0991	31.6520	10
11	1.7103	0.5847	14.2068	0.0704	8.3064	0.1204	4.5144	37.4988	11
12	1.7959	0.5568	15.9171	0.0628	8.8633	0.1128	4.9219	43.6241	12
13	1.8856	0.5303	17.7130	0.0565	9.3936	0.1065	5.3215	49.9879	13
14	1.9799	0.5051	19.5986	0.0510	9.8986	0.1010	5.7133	56.5538	14
15	2.0789	0.4810	21.5786	0.0463	10.3797	0.0963	6.0973	63.2880	15
16	2.1829	0.4581	23.6575	0.0423	10.8378	0.0923	6.4736	70.1597	16
17	2.2920	0.4363	25.8404	0.0387	11.2741	0.0887	6.8423	77.1405	17
18	2.4066	0.4155	28.1324	0.0355	11.6896	0.0855	7.2034	84.2043	18
19	2.5270	0.3957	30.5390	0.0327	12.0853	0.0827	7.5569	91.3275	19
20	2.6533	0.3769	33.0660	0.0302	12.4622	0.0802	7.9030	98.4884	20
21	2.7860	0.3589	35.7193	0.0280	12.8212	0.0780	8.2416	105.6673	21
22	2.9253	0.3418	38.5052	0.0260	13.1630	0.0760	8.5730	112.8461	22
23	3.0715	0.3256	41.4305	0.0241	13.4886	0.0741	8.8971	120.0087	23
24	3.2251	0.3101	44.5020	0.0225	13.7986	0.0725	9.2140	127.1402	24
25	3.3864	0.2953	47.7271	0.0210	14.0939	0.0710	9.5238	134.2275	25
26	3.5557	0.2812	51.1135	0.0196	14.3752	0.0696	9.8266	141.2585	26
27	3.7335	0.2678	54.6691	0.0183	14.6430	0.0683	10.1224	148.2226	27
28	3.9201	0.2551	58.4026	0.0171	14.8981	0.0671	10.4114	155.1101	28
29	4.1161	0.2429	62.3227	0.0160	15.1411	0.0660	10.6936	161.9126	29
30	4.3219	0.2314	66.4388	0.0151	15.3725	0.0651	10.9691	168.6226	30
31	4.5380	0.2204	70.7608	0.0141	15.5928	0.0641	11.2381	175.2333	31
32	4.7649	0.2099	75.2988	0.0133	15.8027	0.0633	11.5005	181.7392	32
33	5.0032	0.1999	80.0638	0.0125	16.0025	0.0625	11.7566	188.1351	33
34	5.2533	0.1904	85.0670	0.0118	16.1929	0.0618	12.0063	194.4168	34
35	5.5160	0.1813	90.3203	0.0111	16.3742	0.0611	12.2498	200.5807	35
40	7.0400	0.1420	120.7998	0.0083	17.1591	0.0583	13.3775	229.5452	40
45	8.9850	0.1113	159.7002	0.0063	17.7741	0.0563	14.3644	255.3145	45
50	11.4674	0.0872	209.3480	0.0048	18.2559	0.0548	15.2233	277.9148	50
55	14.6356	0.0683	272.7126	0.0037	18.6335	0.0537	15.9664	297.5104	55
60	18.6792	0.0535	353.5837	0.0028	18.9293	0.0528	16.6062	314.3432	60
65	23.8399	0.0419	456.7980	0.0022	19.1611	0.0522	17.1541	328.6910	65
70	30.4264	0.0329	588.5285	0.0017	19.3427	0.0517	17.6212	340.8409	70
75	38.8327	0.0258	756.6537	0.0013	19.4850	0.0513	18.0176	351.0721	75
80	49.5614	0.0202	971.2288	0.0010	19.5965	0.0510	18.3526	359.6460	80
85	63.2544	0.0158	1245.0871	0.0008	19.6838	0.0508	18.6346	366.8007	85
90	80.7304	0.0124	1594.6073	0.0006	19.7523	0.0506	18.8712	372.7488	90
95	103.0347	0.0097	2040.6935	0.0005	19.8059	0.0505	19.0689	377.6774	95
100	131.5013	0.0076	2610.0252	0.0004	19.8479	0.0504	19.2337	381.7492	100

© CourseSmart